

Logical & Physical Data Independence

Q. Define Data Independence. Explain the concepts of Logical and Physical Data Independence and highlight the key differences between them.

****Definition to Remember****

****Data Independence****
Data Independence means that the logical structure of the data changes without affecting the physical structure.

1. Types of Data Independence

Data independence is linked to the three-level abstraction architecture and is classified into two types:

1. Physical Data Independence

*** **Concept** ***
Physical Data Independence means that the physical structure of the data changes without affecting the application of the data.

2. Logical Data Independence

*** **Concept** ***
Logical Data Independence means that the logical structure of the data changes without affecting the schema of the data.

2. Difference: Physical vs Logical Data Independence

| Feature | Physical Data Independence | Logical Data Independence |

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****Must-Write Points (for 10 marks)****

1. Data Independence

Quick Recall

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